Q.No 6:

Question description

Selvan studies, engineering as per his father's wishes, while Aaron, whose family is poor, studies engineering to improve his family's financial situation. sumanth, however, studies engineering of his simple passion for developing data structure applications.

sumanth is participating in a hackathon for data structure application development.

sumanth task is to use Insertion Sort to sort the supplied set of numbers.

As a result, The input provides the number of components on the first line and the numbers to be sorted on the second line. Print the array's state at the third iteration and the final sorted array in the supplied format in the output.

Judge will determine whether the outcome is correct or not.

Can you help him ?

Constraints

1 <= N <= 10^5

1 <= Ai <= 10^9

Input Format:

The first line of the input contains the number of elements

the second line of the input contains the numbers to be sorted.

Output Format:

Fist line indicates print the status of the array at the 3rd iteration

second line print the the final sorted array in the given format.

[redacted]

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
64 25 22 90 35
```

EXPECTED OUTPUT

```
22 25 64 90 35
22 25 35 64 90
```

Test Case 2

INPUT (STDIN)

```
5
16 21 22 10 35
```

EXPECTED OUTPUT

```
16 21 22 10 35
10 16 21 22 35
```

Mandatory Test Cases

Code:

```c
#include <stdio.h>

void printArray(int arr[], int n)
```

```
{
    for (int i = 0; i < n; i++)
    {
        printf("%d", arr[i]);

        if (i < n - 1)
            printf(" ");
    }
    printf("\n");
}

void insertionSort(int arr[], int n)
{
    int i, j, key;

    for (i = 1; i < n; i++)
    {
        key = arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
        if (i == 3)
        {
            printArray(arr, n);
        }
    }
}

int main()
{
    int n;
    int arr[100000];

    scanf("%d", &n);
    printf("\noutput:\n ");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    insertionSort(arr, n);
    printArray(arr, n);
```

```
    return 0;
}
```

Output:

Test_case 1:

```
5
64 25 22 90 35
output:

22 25 64 90 35
22 25 35 64 90
```

Test_case 2:

```
5
16 21 22 10 35
output:

10 16 21 22 35
10 16 21 22 35
```